

How to Install MongoDB for This Course

Two supported methods: Installer or Docker

Choose one — both work for the course.

1. Method 1 — Install MongoDB via Installer

This is the simplest option if you want a traditional local installation.

Step 1 — Download MongoDB Community Edition

1. Go to the official MongoDB download page: <https://www.mongodb.com/try/download/community>
2. Choose your operating system: Windows, macOS, or Linux.
3. Select **version 8.3.x** and download the installer.

FIPS note: The community installer does not enable FIPS mode by default. For FIPS-compliant deployments use the Docker method below (UBI9 image) or MongoDB Enterprise.

Step 2 — Run the installer

During installation:

- Accept the default settings.
- Keep the default port **27017**.
- Choose **Complete** installation type.
- Optionally install **MongoDB Compass** if you want a GUI (not required for this course).

Step 3 — Verify installation

1. Open a terminal.
2. Run:

```
mongod --version
```

If you see a version number, MongoDB is installed correctly.

2. Method 2 — Install MongoDB via Docker (recommended)

This is the cleanest option: no system-level installation, easy to reset, and identical across machines. The image used (**mongodb-community-server:8.3-ubi9**) is built on Red Hat Universal Base Image 9 (UBI9).

Step 1 — Install Docker Desktop

1. Download Docker Desktop for your operating system.
2. Install and open it.

3. Make sure Docker is running.

Step 2 — Start MongoDB using Docker

Make sure you have the `docker-compose.yml` file from this folder available. Ideally, clone the entire repository.

Run this command from the directory containing `docker-compose.yml`:

```
docker compose -f docker-compose.yml up -d
```

Or use the provided script:

```
bash run.sh
```

Step 3 — Verify

Run:

```
docker ps
```

You should see a container named `mongodb83`. You can also verify with Docker Desktop's GUI.

Connection details:

Parameter	Value
Host	localhost
Port	27017
Username	mongo
Password	mongo

3. Install the VS Code MongoDB Extension

Use this extension to browse collections and run queries directly inside VS Code.

Step 1 — Open VS Code

Step 2 — Install the extension

1. Open the Extensions panel.
2. Search for **MongoDB for VS Code** by **MongoDB Inc..**
3. Click **Install**.

Step 3 — Connect to your database

1. Click the MongoDB leaf icon in the Activity Bar.
2. Click **Add Connection**.
3. Enter the connection string:

```
mongodb://mongo:mongo@localhost:27017
```

4. Click **Connect**.

You should now see your databases listed in the sidebar.

Step 4 — Run queries

1. Open the MongoDB Playgrounds panel (Ctrl/Cmd + Shift + P → "MongoDB: Create Playground").
2. Write your aggregation pipelines or CRUD operations.
3. Press the Run button or use Ctrl/Cmd + Shift + P → "MongoDB: Run All Playground Blocks".

Results appear inside VS Code.